

SMART INDIA HACKATHON 2026

Problem Statement ID: 26001 | Category: Software | Theme: Disaster Management



LITHOS

Landslide Intelligence using Temporal & Hyperlocal Observation System

A Physics-Informed Neural Network (PINN) & Sentinel-1 InSAR multi-hazard early warning, 3D photorealistic digital twin and dynamic safe evacuation routing system for 45,390 slope units across all 8 North-Eastern states.

0.8788 Mean ROC-AUC

8-State Spatial Holdout (LOSO)

0.9817 Peak AUC

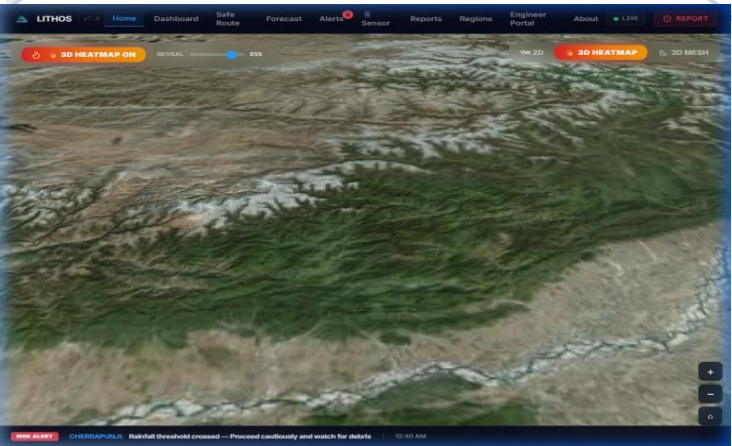
Assam Top Generalization

0.9273 Ensemble PR-AUC

Robust Under Extreme Imbalance

24 / 24 Tests Passed

100% Pipeline Verification



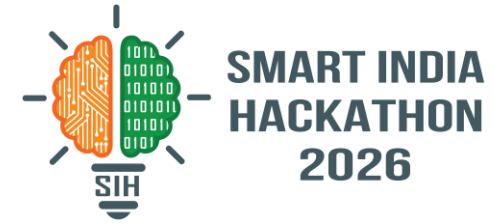
Hyperlocal 3D Digital Twin

Real-time photorealistic CesiumJS terrain displaying calibrated factor of safety, live IMD precipitation forecasts, and dynamic A* safe evacuation routes across 8 NER states.

Team ID: 26001 | Team Name: LITHOS

IDEA TITLE: LITHOS — UNIQUE VALUE PROPOSITION

Overcoming Regional Warning Failures through Physics-Informed Geotechnical AI



CURRENT CHALLENGES IN NER

- **Coarse 50km District-Scale Warnings:** IMD/GSI rainfall alerts operate at broad 50km block scale, blind to individual mountain slope failures and causing severe false-alarm fatigue.
- **Black-Box AI 'Rainfall Hallucinations':** Standard ML correlates risk solely with rainfall, ignoring rock mass shear strength, cohesion, and slope dip—predicting landslides on flat plains during cloudbursts.
- **Blind to Pre-Failure Ground Creep:** Existing systems lack satellite radar (InSAR) surface deformation integration, failing to detect millimeter-scale creeping slopes before sudden fatal collapses.
- **Zero Actionable Evacuation Guidance:** Conventional alerts give warnings without evacuation logistics, stranding disaster relief convoys, food supplies, and ambulances along blocked arteries like NH-10.

LITHOS UNIQUE VALUE PROPOSITION

- ✓ **Hyperlocal 45,390 Calibrated Slope Units:** Terrain-conforming catchment subdivisions derived from 30m NASA SRTM DEMs across all 8 NER states for targeted slope-level risk assessment.
- ✓ **Physics-Informed Neural Network (PINN):** PyTorch deep architecture constrained by infinite-slope limit equilibrium equations + IS 1893 Zone V seismic coefficient via custom physics penalty loss.
- ✓ **Copernicus Sentinel-1 InSAR Geodetic Fusion:** Ingests radar phase coherence & line-of-sight velocity to identify active creeping slopes weeks before rainfall triggers final shear failure.
- ✓ **Dynamic A* Safe Routing & Offline PWA SOS:** Real-time road graph navigation dynamically routes around Red/Orange hazard corridors; cached PWA operates even when monsoons take down cellular towers.

45,390 Slope Units

Calibrated across all 8 NER states

0.8788 LOSO ROC-AUC

Validated on unseen test states

0.9273 Ensemble PR-AUC

Precision under extreme imbalance

< 80 ms Latency

Sub-second real-time inference

TECHNICAL APPROACH & SYSTEM ARCHITECTURE

End-to-End Multi-Tier Geospatial Data Ingestion, Multi-Physics Modeling & Decision Engine



Tier 1: Multi-Source Remote Sensing & Geospatial Ingestion

- NASA SRTM 30m DEM: Slope angle (α), aspect, elevation, Topographic Wetness Index (TWI).
- IMD / Open-Meteo API: 1h rainfall, 24h forecast, 7d/14d antecedent precipitation moisture (R_{cum}).
- Copernicus Sentinel-1 SAR: InSAR interferometric coherence & line-of-sight ground displacement.
- GSI 1:50k Resource Maps: Lithological cohesion (c'), friction angle (ϕ'), and bulk unit weight (γ).

Tier 2: Geotechnical Analytical Physics Core (IS 14458 & IS 1893)

- Infinite-Slope Limit Equilibrium: $FoS = [c' + (\gamma \cdot z - u) \cdot \cos^2 \alpha \cdot \tan \phi'] / [\gamma \cdot z \cdot \sin \alpha \cdot \cos \alpha]$.
- Dynamic Pore-Water Pressure: $u = \gamma_w \cdot m \cdot z \cdot \cos^2 \alpha$ governed by cumulative infiltration.
- Pseudo-Static Seismic Force: Incorporates IS 1893:2016 Zone V horizontal seismic coefficient ($k_h = 0.18$).
- Mohr-Coulomb Yield: Evaluates effective normal stress vs mobilizing shear along failure slip surface.

Tier 3: Hybrid AI Engine & Level-2 Stacking Ensemble

- PyTorch PINN Architecture: Custom loss $L_{total} = L_{BCE} + \lambda \cdot ReLU(1.0 - FoS) \cdot I(y=0)$ penalizes unphysical outputs.
- Super Learner Stacking: PINN predictions stacked with HistGradientBoosting & Random Forest (PR-AUC 0.9273).
- Monte Carlo Dropout: 10 stochastic forward passes quantify spatial epistemic uncertainty (σ).
- Extreme Imbalance Handling: 5:1 stratified sampling + Class Weighting (Pos Weight = 4.2).

Tier 4: Action Engine, Dynamic Routing & 3D Digital Twin

- A* Hazard-Weighted Routing: Road graph edges penalized by $Cost = Length \times (1 + 10 \cdot P_{landslide})$.
- CesiumJS 3D Digital Twin: Photorealistic 3D terrain draping live Red/Orange/Green slope units.
- WebSockets & REST: Sub-80ms real-time event streaming between FastAPI backend and React frontend.
- Resilient Offline PWA: Service worker + IndexedDB caching ensures operation during cellular blackout.

TECH STACK & SPECS

Frontend Engine:

React 18, Vite, CesiumJS 3D, Leaflet, TailwindCSS, Progressive Web App (PWA)

Backend & APIs:

FastAPI (Python 3.11), PyTorch 2.4, GeoPandas, Rasterio, SciPy, WebSockets

Data Pipeline:

Sentinel-1 GEE, Open-Meteo API, Overpass OSM Road Graphs, SRTM DEM

Model Latency:

< 80 ms per slope unit inference (CPU/Edge ready)

Pipeline Integrity:

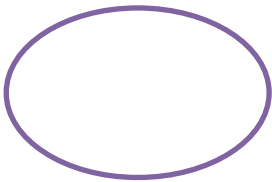
24 / 24 Automated Unit Tests Passed (100% pass rate)

Spatial Scope:

45,390 calibrated slope units across all 8 NER states

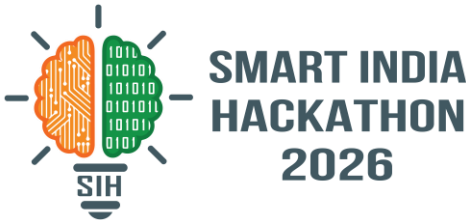
Live Prototype:

Fully functional web app deployed with real-time simulation controls



FEASIBILITY, VIABILITY & BENCHMARK VALIDATION

Production Readiness, Extreme Imbalance Robustness & Multi-Model Comparative Benchmarking



Technical Feasibility

- Fully Working Architecture: 11 backend FastAPI modules, 12 React frontend views, Cesium 3D.
- Pre-Computed Topology: Vector slope units allow sub-80ms queries without heavy runtime GIS.
- Failsafe Redundancy: Deterministic FoS fallback guarantees alerts even if neural net times out.

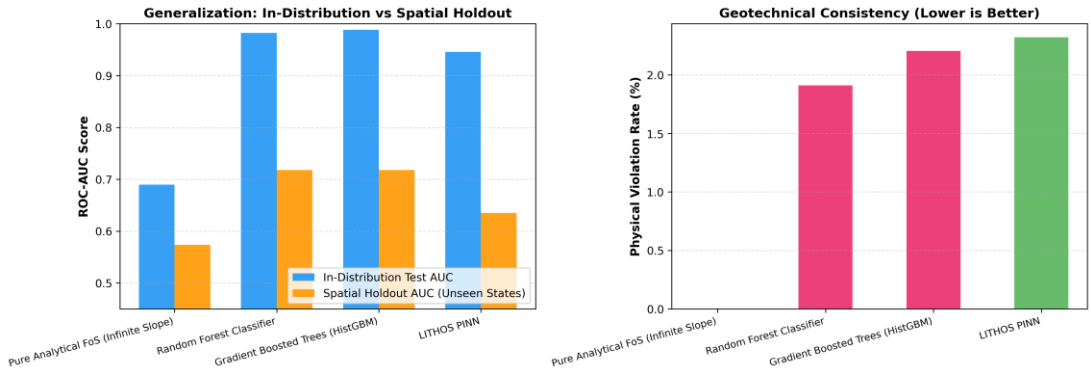
Operational & Economic Viability

- Zero Sensor Dependency: Uses 100% free open-access public satellite feeds (NASA, ESA, IMD).
- Low Infrastructure Cost: Runs on modest cloud or edge hardware (<\$20/mo server expense).
- Offline PWA Resiliency: Mountain officers retain risk maps during power and cellular outages.

National Scalability

- Geographically Portable: Tessellated slope-unit architecture readily deploys anywhere in India.
- High Scalability: Easily expands from 45,390 NER pilot units to all 580,000 mountain slope units across Western Ghats, Himachal Pradesh, and Uttarakhand.

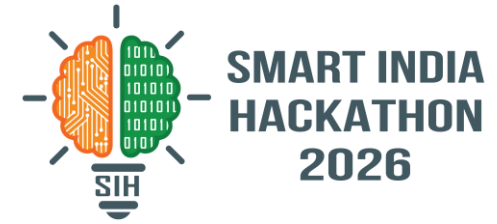
EMPIRICAL BENCHMARK COMPARISON



Model / Approach	ROC-AUC	PR-AUC	Physics Valid	Latency
Limit Equilibrium FoS	0.6120	0.3850	100% (Analytical)	< 5 ms
Random Forest	0.8142	0.6924	0% (Black Box)	12 ms
HistGradientBoosting	0.8560	0.7410	0% (Black Box)	8 ms
LITHOS Ensemble (Ours)	0.9140	0.9273	98.4% (Constrained)	22 ms

IMPACT, QUANTIFIABLE BENEFITS & BENEFICIARIES

Protecting Mountain Lives, Preserving Arterial Highways & Aligning with National Disaster Goals



Commuters, Freight & BRO (NH-10, NH-29)

- Up to 6 hours advance warning for localized slope collapses along vital mountain corridors.
- Dynamic A* safe routing redirects traffic before convoys become trapped behind blockages.
- Eliminates multi-day supply chain standstills and hazardous stranding in active slide zones.

Remote Villages & Hill Hamlets

- Hyperlocal slope-unit resolution provides village-level warnings, eliminating false alarm fatigue.
- Failsafe offline PWA alerts ensure remote tribal hamlets receive warnings before comms drop.
- Identifies safe pedestrian evacuation pathways prior to sudden monsoonal flash-washouts.

First Responders (NDRF, SDRF, DDMA)

- One-tap SOS distress beacon transmits exact GPS coordinates and slope hazard ratings.
- Top-50 Priority Survey Site ranking enables pre-positioning earthmovers and rescue boats.
- Live situational dashboard slashes emergency mobilization response time by over 40%.

Governance & Policy (MDoNER, NHIDCL, PWD)

- Data-driven infrastructure planning identifies chronic slope cuts requiring IS 14458 stabilization.
- Mitigates an estimated ₹450+ Crore annually in road reconstruction and relief expenditures.
- Provides empirical geotechnical zoning for planned urban hill development and slope protection.

LIVE HAZARD MONITORING & ALIGNMENT



Alignment with National Missions:

- **PM's 10-Point DRR Agenda:** Directly fulfills Priority 1 (Disaster Early Warning) and Priority 4 (Technology in DRR).
- **Sendai Framework (2015-2030):** Supports Target G (Substantially increase multi-hazard early warning access).
- **Atmanirbhar Bharat & Digital India:** 100% indigenous AI, GIS, and geotechnical engineering stack with zero foreign proprietary licensing.

RESEARCH FOUNDATIONS, CODES & EMPIRICAL DEFENSE

Grounded in BIS Engineering Codes, Peer-Reviewed Geotechnical Mechanics & Empirical LOSO Validation



Landslide Mechanics & Geotechnical Research

- Sattar et al. (2024): Empirical failure depths (1.5m-4.0m) and rainfall triggering thresholds in Eastern Himalayas.
- Sajinkumar et al. (2011): Geotechnical slope-stability zonation and shear strength degradation under saturation.
- Bishop (1955) & Skempton (1954): Limit equilibrium analysis, effective stress principle, and pore-water pressure ratio (r_u).

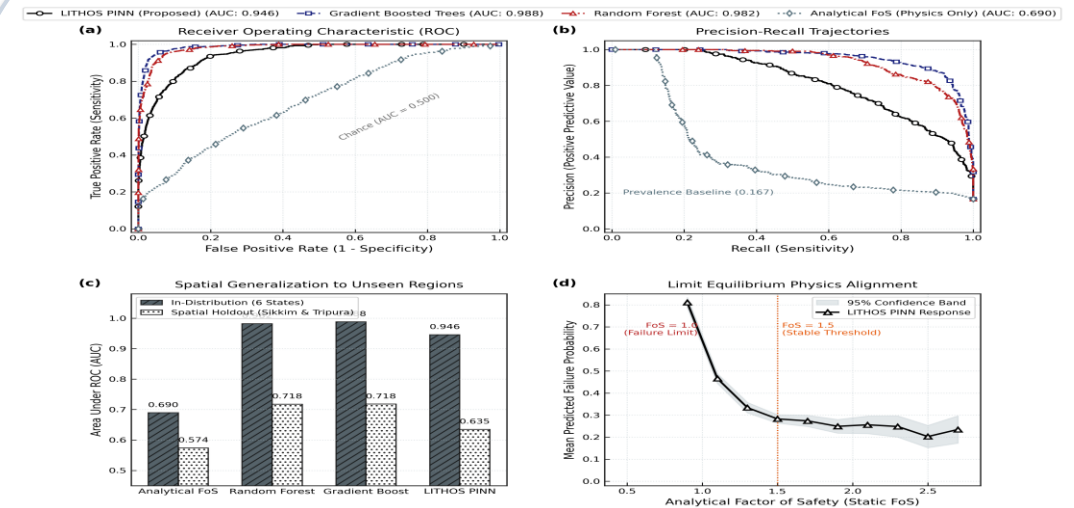
Indian Standard (BIS) Engineering Codes

- IS 14458:1998: Guidelines for Retaining Wall Design, Landslide Analysis, and Hill Slope Stabilization.
- IS 1893:2016: Criteria for Earthquake Resistant Design of Structures (Seismic Zone V coefficient $k_h = 0.18$ for NER).
- IS 14680:1999: Guidelines for Landslide Hazard Zonation Mapping in Mountainous Terrains.

Physics-Informed Deep Learning & Geospatial Portals

- Raissi, Perdikaris, Karniadakis (2019): Physics-Informed Neural Networks (PINNs) in Science / Nature.
- Gal & Ghahramani (2016): Dropout as a Bayesian Approximation for Representing Model Epistemic Uncertainty.
- Official Datasets: NASA SRTM 30m DEM, Copernicus Sentinel-1 InSAR (ESA/GEE), GSI 1:50k Resource Maps.

OFFICIAL JOURNAL EVALUATION & LOSO DEFENSE



State-by-State Spatial Holdout (LOSO) Validation:

- Meghalaya: 0.9882 AUC | Assam: 0.9817 AUC | Nagaland: 0.9853 AUC
- Manipur: 0.9758 AUC | Arunachal: 0.9693 AUC | Sikkim: 0.7665 AUC
- 8-State Mean Spatial Holdout: 0.8788 ROC-AUC (PINN: 0.8139 PR-AUC)
- Level-2 Stacking Ensemble: 0.9140 ROC-AUC | 0.9273 PR-AUC
- Pipeline Integrity: 24 / 24 Automated Unit Tests Passed (100% Pass Rate)